

Covert, Anonymous and Video-Bound Provenance: Hiding zk-SNARK Camera Proofs in H.264 CAVLC Trailing-One Signs

Author Name(s) (to be filled in)

Affiliation

City, Country

e-mail

Abstract—Deployed content-provenance systems attach a signed manifest to a media file. The manifest is metadata that platforms strip, it identifies the signing device, and its presence is visible. We hide a zero-knowledge proof of origin inside the H.264 bitstream instead. A camera whose key is a leaf of a public Poseidon Merkle registry produces a 129-byte Groth16 proof that it is registered and vouches for a binding of the video and a message; the proof is embedded, under a separate stego key, into CAVLC trailing-one signs of IDR frames, which keeps every codeword length and the file size. To bind the proof to the stream that carries it, we hash every NAL unit with exactly the keyed carrier bits cleared: the digest is invariant under embedding, while any other edit—another video, truncation, re-encoding, an edited non-carrier bit—changes it. Verifiers need the stego key—or a per-video token that exposes nothing else—and the registry root but no camera secret, cannot forge proofs and do not learn which camera signed. We also derive a closed-form distortion model: one sign flip costs $4Q_{\text{step}}^2\kappa$ in pixel energy, $\kappa \in [0.925, 1.057]$, times a propagation gain G ; it inverts into a per-frame bit budget and yields a drift-tier selection policy. The direct energy matches the decoder exactly and G does not depend on QP. On 8 CIF and 19 native 720p/1080p sequences the drift-tier policy gains 3.3–8.0 dB of PSNR (median 4.9 dB on CIF, 5.9 dB on high resolution) over a uniform keyed schedule and outperforms four re-implemented published trailing-one methods at every QP and GOP; four steganalysers (compressed-domain, SPAM, rich model, CNN) are at chance at the default rate of 64 bits per IDR and succeed at 20% of the candidates. Proving takes 1.5–1.8 s, verification about 1 s, and an 11-case attack matrix including replay is rejected as expected.

Index Terms—Video steganography, H.264/AVC, CAVLC, zk-SNARK, Groth16, content provenance, anonymous attestation, distortion model.

I. INTRODUCTION

Whether a video was really captured by a trustworthy device has become a practical question for newsrooms, courts and platforms. The industry answer is the C2PA standard [1]: the capture device signs a manifest that binds a hash of the asset to assertions about its origin, and the manifest travels with the file as metadata. This design has three properties that are undesirable in some settings. First, the manifest is *separable*: social networks and many transcoding pipelines strip metadata, which is why C2PA added watermark-based “soft bindings” to recover a lost manifest. Second, it is *identifying*: the signature is verified against a device or vendor certificate, so the verifier learns who signed. Third, it is *visible*: anyone can see that

the file carries provenance, and an adversary who wants to suppress it knows exactly which files to target.

Zero-knowledge proofs address the second point. A camera can prove that it belongs to a set of authorized devices without revealing which one, the pattern behind anonymous credentials and Semaphore-style group membership [2], [3]. Recent work applies zk-SNARKs to media provenance, mostly to prove that a published image or video is a permissible transformation of a signed original [4]–[9]. Steganography addresses the first and third points: a message hidden in the coded bitstream survives metadata stripping and container changes, and is invisible to a party that does not hold the key [10]–[12]. Compressed-domain data hiding in H.264 has a long history, including embedding in the sign bits of CAVLC trailing ones, which leaves the bit rate unchanged [13]–[15].

This paper combines the two. The question we answer is: *can a short zero-knowledge proof of anonymous camera membership be hidden inside the H.264 bitstream it vouches for, bound to that exact bitstream, at a quality cost that can be predicted and controlled?* Three technical obstacles stand in the way. (i) A proof embedded into a video changes the video, so a proof that commits to the video’s hash cannot be computed before embedding; the binding has to be defined over a canonical form that embedding leaves unchanged, without opening a gap an attacker can exploit. (ii) The proof must be small enough for a covert channel whose capacity is a few dozen bits per frame, and its verifier must not need the camera’s secret—otherwise any verifier could forge proofs, and the proof would add nothing over a shared-key MAC. (iii) Sign flips in intra-coded frames propagate through intra prediction, so the quality cost is not simply proportional to the number of embedded bits; it must be modeled before an embedding budget can be chosen.

Contributions.

- A *video-bound anonymous camera proof* (Section V): a Circom circuit of 8,737 constraints with three public inputs (registry root and a 256-bit binding split in two halves) proving that Poseidon(s) is a leaf of a depth-16 Poseidon Merkle registry, with Groth16 proofs of 129 bytes.
- A *carrier-masked video digest*: SHA-256 over the RBSP of every NAL unit with exactly the keyed carrier bits

cleared. We prove that it is invariant under embedding and that every other edit is detected, either through the digest or through the extracted proof input (Section VI).

- A *keyed, segment-wise covert channel* on CAVLC trailing-one signs with HKDF-derived schedule and whitening keys, a three-byte frame, and one segment per IDR, which works identically on files and on live streams; optional *per-video keys* yield a 64-byte verification token that makes one video publicly verifiable without exposing the stego key or any other video.
- A *closed-form distortion model* of trailing-one sign embedding, derived from the H.264 dequantization and inverse transform, together with its inverse: the number of flips, and hence the per-IDR bit cap, that keeps a frame above a target PSNR (Section VII); and a *drift-tier selection policy* derived from it.
- An implementation and an evaluation (Sections VIII–IX): validation of the model on single-flip and sequence experiments; a comparison with four re-implemented published trailing-one methods on 8 CIF and 19 native 720p/1080p sequences, four QPs and two GOP structures; steganalysis with compressed-domain, rich-model and CNN detectors at three rates; Groth16/PLONK costs; and an 11-case attack matrix.

Scope. The scheme is *fragile* by design: it protects bitstreams that are transferred unchanged (evidence chains, archives, files sent as files). A re-encode destroys both the payload and the digest; robustness to transformations is the subject of the transformation-proof literature [4], [6], [8] and is outside this paper. The proof states that a registered camera vouched for a bitstream, not that the scene is real.

II. RELATED WORK

Data hiding in the H.264 compressed domain. Embedding directly in entropy-coded syntax avoids a full decode–encode cycle. Kim *et al.* [16] write a bit into the sign of the highest-frequency trailing one of luma blocks, which keeps every codeword length and therefore the bit rate; Lin *et al.* [17] embed the XOR of all trailing-one signs of keyed blocks; Wang and Ma [18] use one host block out of every E candidates and a $(1, 3, 2)$ matrix code when a block has three trailing ones; Liao *et al.* [19], [20] hide one bit per intra macroblock in a trailing-one parity. Other CAVLC methods substitute codewords [14] or modify levels [15]. Because intra prediction reuses reconstructed neighbors, such changes *drift* inside the frame; Ma *et al.* [21] choose blocks and modify coefficient pairs so that the error cannot propagate (their conditions are tabulated in [22]). You *et al.* [23] show that CAVLC hiding is detectable and propose a detection strategy. We use the trailing-one sign carrier, add a keyed schedule and whitening, quantify drift with an explicit model, and compare against the sign-reducible methods above under identical conditions (Section IX).

Fragile watermarking and self-authentication. A classic construction hashes the content without the embedding region, signs the hash and embeds the signature into that region [24],

[25]. Our video digest follows this pattern, with two differences that matter for security: the excluded region is *only* the set of keyed carrier positions (not a whole bit plane or block class), and the embedded object is a zero-knowledge proof rather than a signature, so verification reveals neither the signer nor a verification key per device.

Media provenance. C2PA [1] binds signed manifests to assets through hard bindings (hashes) and soft bindings (watermarks or fingerprints). It offers public verifiability and an established certificate infrastructure, but the manifest is separable metadata and the signer is identified. Our proof travels inside the bitstream, is hidden from parties without the stego key, and hides the signer within the registry.

Zero-knowledge proofs for media. PhotoProof [4] proves that an image is derived from a signed original by permissible transformations. ZK-IMG [5] attests images from attested cameras and hides the pre-image of transformations; VerITAS [6] and VIMz [7] scale such proofs to high-resolution images, the latter with folding-based SNARKs and signer anonymity; zk-Cinema [8] extends transformation proofs to video; AIPA [9] adds unlinkable pseudonym certificates. These systems prove *what happened to* a signed asset and keep the proof beside it. We solve a different problem: the proof is hidden *inside* the asset, bound to the exact bitstream, and the signer is anonymous within a registry; we do not support transformations.

Zero knowledge and information hiding. Adelsbach and Sadeghi [26] prove the presence of a watermark in zero knowledge; Toso and Chung [27] combine steganography and an interactive zero-knowledge proof to embed a signature in an image. We embed a non-interactive succinct argument whose statement depends on the cover itself.

Anonymous membership. Proving that a committed key is a leaf of a public Merkle tree [28] without revealing the leaf is the core of Zerocash [2] and Semaphore [3]; Poseidon [29] makes the in-circuit hashing cheap. We use this pattern for a camera registry.

Minimal-distortion steganography. Matrix embedding [30] and syndrome-trellis codes [31], [32] minimize an additive distortion for a given payload. Our model supplies the per-carrier cost such codes need in the CAVLC sign domain.

Table I summarizes the comparison.

III. BACKGROUND

We summarize the parts of H.264/AVC [33]–[35] that the channel and the distortion model rely on, and the zero-knowledge tools of the proof. We restrict attention to the Constrained Baseline profile: progressive 4:2:0, 8-bit, CAVLC entropy coding, flat scaling matrices.

A. Intra coding of an IDR picture

An IDR picture is split into 16×16 macroblocks (MBs) coded in raster order. In an I-slice an MB is either *Intra4×4* (sixteen 4×4 luma blocks, each predicted with one of nine directional modes from reconstructed samples above and to

TABLE I
COMPARISON WITH PROVENANCE AND COMPRESSED-DOMAIN HIDING APPROACHES. ✓: SUPPORTED; -: NOT SUPPORTED; (✓): PARTIALLY.

Approach	In-band (no metadata)	Hidden	Signer anonymous	Bound to exact stream	Survives re-encode	Public verification
C2PA manifest [1]	-	-	-	✓	(✓) ^a	✓
CAVLC fragile watermark [14], [15]	✓	(✓)	-	✓	-	-
CAVLC sign/parity hiding [16]–[18], [20]	✓	✓	-	-	-	-
Signed digest in content [25]	✓	-	-	✓	-	✓
PhotoProof / ZK-IMG [4], [5]	-	-	(✓)	-	✓ ^b	✓
VIMz / zk-Cinema / AIPA [7]–[9]	-	-	✓	-	✓ ^b	✓
This work	✓	✓	✓	✓	-	(✓) ^c

^a via soft-binding watermarks that recover the manifest. ^b for the transformations the proof system supports. ^c per video, through a disclosed verification token (Section V-H); otherwise verifiers need the stego key.

the left), *Intra16×16* (one of four modes for the whole MB: vertical, horizontal, DC, plane), or I_PCM. Chroma (8×8 per MB for 4:2:0) uses one of four modes. The encoder codes the prediction residual $e = x - p$; the decoder reconstructs $\hat{x} = \text{clip}(p + \hat{e})$. Because p is computed from *reconstructed* neighbors, a change in $\hat{e}$ of one block alters the prediction of later blocks: this is intra drift.

B. From the DCT to the integer transform

The 4×4 two-dimensional DCT-II is $Y = AXA^\top$ with

$$A = \begin{bmatrix} a & a & a & a \\ b & c & -c & -b \\ a & -a & -a & a \\ c & -b & b & -c \end{bmatrix}, \quad \begin{aligned} a &= \frac{1}{2}, \\ b &= \frac{1}{\sqrt{2}} \cos \frac{\pi}{8}, \\ c &= \frac{1}{\sqrt{2}} \cos \frac{3\pi}{8}. \end{aligned} \quad (1)$$

H.264 replaces $c/b \approx 0.414$ by $1/2$ and rescales $b = \sqrt{2/5}$ to keep the rows orthogonal [36]. The transform factors into an integer core and a scaling that is folded into quantization:

$$Y = (C_f X C_f^\top) \odot E_f, \quad C_f = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -1 & -2 \\ 1 & -1 & -1 & 1 \\ 1 & -2 & 2 & -1 \end{bmatrix}, \quad (2)$$

where E_f has entries $a^2, ab/2, b^2/4$ according to the parity class of (i, j) : both indices even, both odd, or mixed. The inverse is

$$X = C_i^\top (Y \odot E_i) C_i, \quad C_i = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \frac{1}{2} & -\frac{1}{2} & -1 \\ 1 & -1 & -1 & 1 \\ \frac{1}{2} & -1 & 1 & -\frac{1}{2} \end{bmatrix}, \quad (3)$$

with E_i entries a^2, ab, b^2 . Only additions and shifts are needed, and encoder and decoder match exactly. We write $b_0, \dots, b_3$ for the rows of C_i ; their squared norms are $\|b_0\|^2 = \|b_2\|^2 = 4$ and $\|b_1\|^2 = \|b_3\|^2 = 5/2$.

C. Quantization and scaling

The quantizer step Q_{step} doubles every six QP values:

$$Q_{\text{step}}(QP) = Q_{\text{step}}(QP \bmod 6) \cdot 2^{\lfloor QP/6 \rfloor}, \quad (4)$$

with $Q_{\text{step}}(0..5) = 0.625, 0.6875, 0.8125, 0.875, 1, 1.125$. The encoder computes $W = C_f X C_f^\top$ and quantizes

$$|Z_{ij}| = (|W_{ij}| MF_{ij} + f) \gg q_{\text{bits}}, \quad q_{\text{bits}} = 15 + \lfloor QP/6 \rfloor, \quad (5)$$

with $MF = 2^{q_{\text{bits}}} PF / Q_{\text{step}}$, $PF \in \{a^2, ab/2, b^2/4\}$, and a dead-zone offset f ($2^{q_{\text{bits}}}/3$ for intra in the reference design; x264 additionally applies trellis and psychovisual decisions). The decoder scales a level c_{ij} by

$$d_{ij} = c_{ij} \cdot v_m(i, j) \cdot 2^{\lfloor QP/6 \rfloor}, \quad m = QP \bmod 6, \quad (6)$$

where $v_m(i, j) \in \{v_{m,0}, v_{m,1}, v_{m,2}\}$ by parity class (even/even, odd/odd, mixed):

m	0	1	2	3	4	5
$v_{m,0}$	10	11	13	14	16	18
$v_{m,1}$	16	18	20	23	25	29
$v_{m,2}$	13	14	16	18	20	23

(7)

(The standard writes $\text{LevelScale} = 16 v_m$ and a shift by four for flat matrices; the product is the same.) The residual is then $\hat{e} = (C_i^\top D C_i + 32) \gg 6$, where C_i is applied with integer shifts. Note that $Q_{\text{step}}(m) = v_{m,0}/16$: the step size is the even/even scaling factor.

DC paths. In an Intra16×16 MB the sixteen luma DC coefficients form a 4×4 block coded separately (category *LumaDC*), transformed by a 4×4 Hadamard matrix before scaling; the remaining fifteen coefficients of each block are *Luma AC*. For chroma, the four DC coefficients of each 8×8 component form a 2×2 block (*ChromaDC*) with a 2×2 Hadamard transform, and each 4×4 block has fifteen *ChromaAC* coefficients. Chroma uses QP_c , derived from $QP + \text{chroma_qp_index_offset}$ by a table that saturates at 39.

D. CAVLC

After a zig-zag scan, each block of levels is coded as [33], [35]:

- 1) `coeff_token`: the pair (TotalCoeff, TrailingOnes), from one of four VLC tables chosen by n_C , the rounded mean of TotalCoeff of the left and upper blocks;

- 2) one `trailing_ones_sign_flag` per trailing one—the up to three highest-frequency non-zero coefficients equal to ± 1 —in reverse scan order, $0 \mapsto +1, 1 \mapsto -1$;
- 3) the remaining levels (`level_prefix`, `level_suffix`, `adaptive suffixLength`), `total_zeros` and `run_before`.

Proposition 1 (Syntax invariance). *Flipping a `trailing_ones_sign_flag` changes exactly one level from ± 1 to ∓ 1 and leaves the length and the decoded value of every other syntax element of the slice unchanged.*

Proof. The flag is a fixed one-bit field. `TotalCoeff` and `TrailingOnes`, hence the `coeff_token` codeword and the n_C context of later blocks, do not depend on signs. The initial `suffixLength` depends only on `TotalCoeff` and `TrailingOnes`, its updates only on the non-trailing-one levels, and the magnitude adjustment of the first non-trailing-one level only on `TrailingOnes`. `total_zeros` and `run_before` depend on positions only. Hence every later bit is parsed at the same offset with the same meaning. $\square$

Consequently the edited stream is a valid stream of the same RBSP length, and the set of trailing-one sign positions (the *candidates*) is the same in the cover and the stego stream. We use the *first* sign flag of each block that has at least one trailing one, i.e. the highest-frequency trailing one.

E. NAL units and emulation prevention

An Annex-B stream is a sequence of NAL units separated by start codes `00 00 01`. Each NAL unit has a one-byte header (`forbidden_zero_bit`, `nal_ref_idc`, `nal_unit_type`) followed by the EBSP: the RBSP with an emulation-prevention byte `03` inserted after every `00 00` that precedes a byte ≤ 03 . Bit offsets in this paper are RBSP offsets. A sign flip can add or remove one emulation-prevention byte, which changes the file size by one byte but no RBSP offset.

F. zk-SNARKs, Groth16 and Poseidon

A statement “ $\exists w : \mathcal{R}(x, w)$ ” is compiled to a rank-1 constraint system $(Az) \circ (Bz) = Cz$ over the BN254 scalar field [37], $z = (1, x, w)$. Groth16 [38] proves satisfiability with $\pi = (A, B, C) \in \mathbb{G}_1 \times \mathbb{G}_2 \times \mathbb{G}_1$, verified with one pairing-product equation; it is complete, knowledge-sound and perfectly zero-knowledge in the generic group model. The common reference string comes from a universal Phase 1 (Powers of Tau) and a circuit-specific Phase 2; the multi-party ceremony is secure if one participant is honest [39]. A proof compresses to 129 bytes: the x -coordinates of A , B (two $\mathbb{F}_q$ elements) and C , plus one byte of sign flags. PLONK [40] trades a larger proof for a universal setup. Poseidon [29] is a sponge hash over the scalar field designed for few constraints; we use it for public keys and Merkle nodes. Circuits are written in Circom [41] and proven with snarkjs [42].

IV. SYSTEM AND THREAT MODEL

Parties. A *registry authority* publishes the root ρ of a Merkle tree of camera public keys. A *camera* holds a secret s with Poseidon(s) in the tree. A *sender* (the camera or an embedder it trusts) holds a 32-byte stego key K shared with the intended verifiers, and turns a cover H.264 stream into a stego stream carrying a message m and a proof. The *channel* delivers the bitstream, possibly through a party that inspects or edits it. A *verifier* knows K , ρ and the Groth16 verification key, and accepts or rejects.

Two secrets, two roles. K hides and locates the payload; s authorizes it. Holding K lets a verifier extract and check a proof but not create one, which is the point of using a zero-knowledge proof instead of a MAC under K .

Adversary. The adversary sees stego streams and may know K (a curious or malicious verifier) or not (a channel observer). It may edit, truncate, splice or re-encode streams, move a valid payload into another stream, change the message, register its own cameras in a tree of its choice, and obtain proofs from honest cameras for streams of its choice. It does not know the secret of any honest registered camera and cannot break SHA-256, Poseidon, HMAC-SHA256 or the Groth16 assumptions.

Goals.

- G1 *Validity*: an honest stego stream is a valid H.264 stream of the same syntax as the cover.
- G2 *Binding*: if the verifier accepts a stream V' with message m' , then a registered camera produced a proof for exactly (V', m') up to the carrier bits that hold the payload.
- G3 *Unforgeability*: without the secret of a registered camera, no adversary—including one holding K —makes the verifier accept a (stream, message) pair that no registered camera vouched for.
- G4 *Anonymity*: the verifier learns nothing about which registered camera produced the proof.
- G5 *Covertiness (empirical)*: a party without K cannot locate the payload, and first-order sign statistics do not reveal embedding.

Non-goals. Robustness to transcoding or editing; proving that the camera recorded a real scene (a camera filming a screen produces a valid proof); verification without K ; resistance to modern learned steganalysis, which we have not evaluated; protection of the camera secret itself, which in our prototype lives in software.

V. DESIGN

Fig. 1 shows the data flow. The sender computes the video digest of the cover under K , binds it with the message, obtains a proof from the camera, packs message and proof into a payload, frames it and embeds the frame into the trailing-one signs selected by a keyed schedule. The verifier reverses the channel, recomputes the digest from the stream it received and verifies the proof against the registry root.

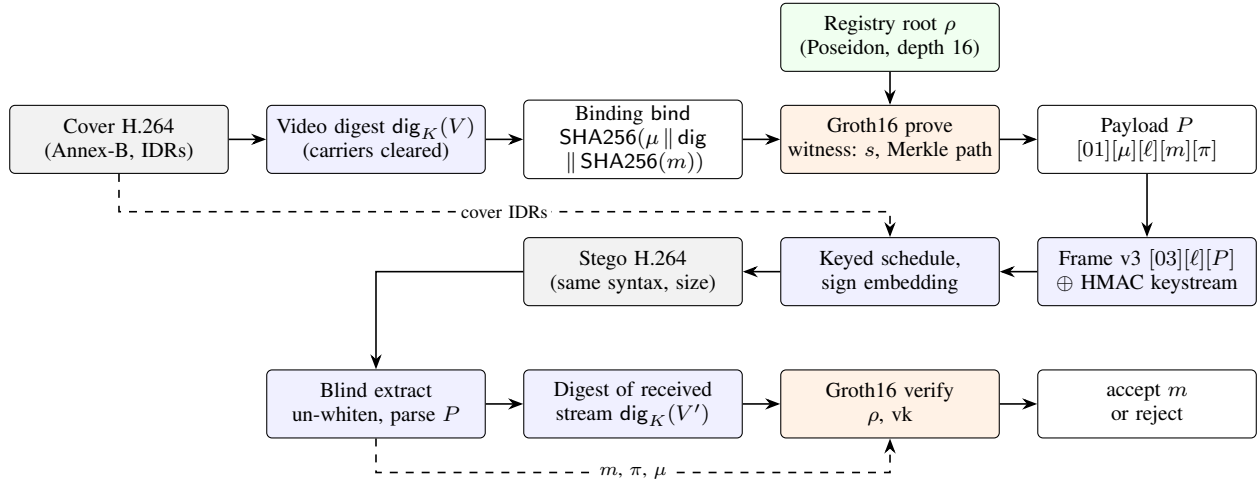


Fig. 1. Data flow. Blue: covert channel under the stego key K ; orange: zero-knowledge proof; the verifier never sees s .

A. Keys

From the 32-byte stego key K the channel derives two subkeys with HKDF-SHA256 [43]:

$$PRK = \text{HMAC}(\text{"zkstego-cavlc-v2-salt"}, K), \quad (8)$$

$$k_{\text{sch}} = \text{Expand}(PRK, \text{".../schedule"}, 32), \quad (9)$$

$$k_{\text{wh}} = \text{Expand}(PRK, \text{".../whitening"}, 32). \quad (10)$$

The camera secret s is an independent uniformly random non-zero element of the BN254 scalar field; its public key is $pk = \text{Poseidon}(s)$.

B. Candidates and segments

Each IDR picture together with its preceding SPS and PPS forms a *segment*. A native parser decodes the CAVLC syntax of the IDR slice and lists one candidate per 4×4 (or 2×2 chroma DC) block that has a trailing one: the RBSP offset of its first `trailing_ones_sign_flag`. A candidate is identified by the string `id = nal:mb:category:block:bit` of its analysis NAL index, MB address, category (LumaDC, Luma4x4/AC, ChromaDC, ChromaAC), block index and RBSP bit offset. By Proposition 1 the identifiers are the same before and after embedding.

C. Keyed schedule

Within segment j with candidate set $\mathcal{C}_j$, every candidate gets the score $\sigma(id) = \text{HMAC}(k_{\text{sch}}, id)$; candidates are sorted by (σ, id) and the first $n_j = \min(\text{cap}, |\mathcal{C}_j|)$ are selected. Frame bits are assigned to segments in order: segment j carries frame bits $[\sum_{i < j} n_i, \sum_{i \leq j} n_i]$ until the frame ends. The schedule depends only on K , the stream syntax and the per-IDR cap (64 bits by default), so a receiver reconstructs it from the stego stream alone, and the same procedure runs on a live stream as each IDR arrives.

D. Frame and whitening

The channel frame (protocol version 3) is

$$F = 0 \times 03 \parallel \text{u16}(|P|) \parallel P, \quad (11)$$

with no MAC. Bit i of F (MSB first, counted across segments) is XORed with bit i of the keystream $\text{HMAC}(k_{\text{wh}}, \text{u64}(0)) \parallel \text{HMAC}(k_{\text{wh}}, \text{u64}(1)) \parallel \dots$ and written as the sign of the i -th scheduled candidate. Whitening makes the embedded bits pseudo-random, so about half of the scheduled signs actually change, and a wrong key yields a random version byte and length, which the extractor rejects. Authentication of the payload is left entirely to the proof; an earlier version carried a 16-byte HMAC tag, which cost 128 bits (two IDRs at the default cap) and added nothing once the proof binds the message.

E. Camera registry

The registry is a binary Merkle tree of depth 16 whose leaves are camera public keys (unused leaves are 0) and whose internal nodes are $\text{Poseidon}(\ell, r)$. Verifiers trust only the root ρ ; a camera's membership witness is its sibling path $(\text{sib}_t, \beta_t)_{t < 16}$.

F. Carrier-masked video digest

Let $\Gamma_K(V)$ be the set of (NAL, RBSP bit) positions that the schedule of K selects to carry the $L_F = 8|F|$ frame bits of a payload of the given length. With the domain string $\text{dom}_v = \text{"zkstego/video-digest/v1"}$, the digest is

$$\text{dig}_K(V) = \text{SHA256}(\text{dom}_v \parallel \text{u32}(L_F) \parallel \text{u32}(\text{cap}) \parallel \parallel_{u \in V} \text{u32}(1 + |r'_u|) \parallel h_u \parallel r'_u), \quad (12)$$

where for each NAL unit u in file order h_u is its header byte and r'_u its RBSP with every bit in $\Gamma_K(V)$ cleared. The schedule depends on the syntax only, so $\Gamma_K(V) = \Gamma_K(V^*)$ for a cover V and its stego V^* .

G. Binding, statement and payload

For a mode $\mu \in \{0, 1\}$ (0: video-bound; 1: message-only, Section X) the binding, with $\text{dom}_b = \text{"zkstego/proof-binding/v1"}$, is

$$\text{bind} = \text{SHA256}(\text{dom}_b \parallel \mu \parallel \text{dig}_K(V) \parallel \text{SHA256}(m)), \quad (13)$$

split into two 128-bit field elements $(\text{bind}_{\text{hi}}, \text{bind}_{\text{lo}})$. The circuit proves

$$\mathcal{R} = \{(\rho, \text{bind}_{\text{hi}}, \text{bind}_{\text{lo}}; s, \text{sib}, \beta) :$$

$$\text{MerkleRoot}(\text{Poseidon}(s), \text{sib}, \beta) = \rho \wedge \beta_t \in \{0, 1\}\}. \quad (14)$$

The binding halves are public inputs that enter the R1CS through a squaring constraint, so a proof verifies only for the binding it was produced for. The circuit has 8,737 constraints and three public inputs. The payload is

$$P = 0x01 \parallel \mu \parallel \text{u16}(|m|) \parallel m \parallel \pi, \quad |\pi| = 129, \quad (15)$$

so $L_F = 8(|m| + 136)$; a 68-byte message gives $|P| = 201$, $|F| = 204$ bytes, 1,632 bits, i.e. 26 IDRs at 64 bits each.

H. Channel parameters: selection policy and key mode

Two optional parameters are agreed out of band, like the cap; their defaults reproduce the channel described so far byte for byte.

Selection policy. Policy random ranks a segment's candidates by (σ, id) . Policy low-drift ranks them by (τ, σ, id) , where the integer *drift tier*

$$\tau = r + d + f \in \{0, \dots, 5\} \quad (16)$$

is derived from the distortion model of Section VII: $r = 2, 1, 0$ for an MB row in the top, middle or bottom third of the picture (more blocks predict from upper rows), $d = 2$ for the DC paths LumaDC and ChromaDC (their change covers a whole MB component), and $f = 1$ when the flipped AC coefficient has $i+j \leq 2$ (low frequencies change the right column and bottom row used for prediction). All three inputs are syntax that sign flips never change, so the receiver recomputes the tiers from the stego stream; inside a tier the order stays keyed.

Per-video keys and verification tokens. In key mode per-video the subkeys are bound to the video. With $\nu = \text{SHA256}(\text{"zkstego/video-nonce/v1"} \parallel \dots)$ computed like Eq. (12) over the SPS, PPS and first IDR of the stream with *all* trailing-one sign candidates cleared (so ν is the same for cover and stego and independent of any key),

$$\begin{aligned} PRK' &= \text{Extract}(\text{"...-v4-salt"}, K), \\ k_v &= \text{Expand}(PRK', \text{".../video"} \parallel \nu, 32), \\ k_{\text{sch}} &= \text{Expand}(k_v, \text{".../schedule"}, 32), \\ k_{\text{wh}} &= \text{Expand}(k_v, \text{".../whitening"}, 32). \end{aligned} \quad (17)$$

The pair $t_V = (k_{\text{sch}}, k_{\text{wh}})$ is the *verification token* of video V . Anyone given t_V can extract the payload of V and compute dig of V (Algorithm 2 with t_V in place of K) and so verify its proof publicly, without learning K or the token of any other video. A live stream obtains ν from its first segment as it arrives. The verify job accepts a token in place of the key in this mode.

Algorithm 1 Embed (sender)

Require: cover V , message m , stego key K , camera (s, path) , cap

- 1: $L_F \leftarrow 8(|m| + 136)$; $\Gamma \leftarrow \text{schedule}(V, K, L_F, \text{cap})$
- 2: $\delta \leftarrow \text{dig}_K(V)$ $\triangleright$ Eq. (12)
- 3: $\text{bind} \leftarrow \text{SHA256}(\text{dom} \parallel 0 \parallel \delta \parallel \text{SHA256}(m))$
- 4: $\pi \leftarrow \text{Groth16.Prove}(\rho, \text{bind}; s, \text{path})$
- 5: $F \leftarrow 03 \parallel \text{u16}(|P|) \parallel P$ with $P = 01 \parallel 0 \parallel \text{u16}(|m|) \parallel m \parallel \pi$
- 6: **for** $i = 0$ **to** $L_F - 1$ **do**
- 7: set the sign at $\Gamma[i]$ to $F_i \oplus ks_i$
- 8: **end for**
- 9: **return** V^*

Algorithm 2 Verify (receiver)

Require: stream V' , K , root ρ , verification key vk , cap

- 1: read the first 24 scheduled bits, un-whiten; **reject** unless version = 3 and length is admissible
- 2: read and un-whiten the remaining bits; parse P ; **reject** if malformed
- 3: **if** $\mu = 1$ and message-only proofs are not allowed **then reject**
- 4: **end if**
- 5: $\delta' \leftarrow \text{dig}_K(V')$ with L_F from the header
- 6: $\text{bind}' \leftarrow \text{SHA256}(\text{dom} \parallel \mu \parallel \delta' \parallel \text{SHA256}(m))$
- 7: **return** $\text{Groth16.Verify}(\text{vk}, (\rho, \text{bind}'_{\text{hi}}, \text{bind}'_{\text{lo}}), \pi)$

I. Algorithms

Algorithms 1 and 2 summarize sender and receiver. The service exposes the verifier as an asynchronous job whose outcomes are `payload_not_found`, `malformed_proof_payload`, `video_binding_required`, `video_digest_unavailable`, `proof_invalid` or acceptance with `video_bound`; an extraction-only job returns the payload marked `verified: false`.

VI. SECURITY ANALYSIS

We argue goals G1–G5 informally under standard assumptions: SHA-256 and the Poseidon Merkle tree are collision resistant, Poseidon is one-way on uniformly random inputs, HMAC-SHA256 is a PRF, and Groth16 is perfectly zero-knowledge and weakly simulation-extractable: an adversary that has seen proofs for some public inputs can re-randomize them, but cannot produce an accepting proof for a *new* public input without knowing a witness [44].

Proposition 2 (Validity, G1). *Embedding changes only bits in $\Gamma_K(V)$; the stego stream parses into the same syntax elements with the same lengths as the cover, and only the selected trailing-one levels change sign.*

Proof. Every bit in $\Gamma_K(V)$ is a `trailing_ones_sign_flag`; apply Proposition 1 once per flip, in any order. $\square$

Lemma 1 (Digest invariance). *For a cover V and its stego stream V^* under key K , $\text{dig}_K(V^*) = \text{dig}_K(V)$.*

Proof. By Proposition 2 both streams have the same NAL units, headers and RBSP lengths (emulation-prevention bytes are not part of the RBSP), and the same candidates, so the schedule yields $\Gamma_K(V^*) = \Gamma_K(V)$. Their RBSPs differ only inside this set, which Eq. (12) clears in both. $\square$

Lemma 2 (Syntax determines the mask). *If $\text{dig}_K(V') = \text{dig}_K(V)$ for a stream V' , then, except with negligible probability, V' and V have the same NAL units and RBSP lengths and differ only at positions in $\Gamma_K(V) = \Gamma_K(V')$.*

Proof. Collision resistance of SHA-256 gives equal canonical strings, hence equal headers and lengths, and RBSPs that agree outside $\Gamma_K(V) \cup \Gamma_K(V')$. Consider the first differing bit in parsing order. Both streams parse identically up to it, so it has the same syntactic role in both; it lies in one of the two masks, hence it is a trailing-one sign flag in both. By Proposition 1 the parse of the rest is unaffected, and induction over the differing bits shows that V and V' have identical syntax and candidates. The schedule depends only on syntax and K , so the masks are equal. $\square$

Theorem 1 (Binding, G2). *Suppose the verifier accepts (V', m') in mode 0 and the adversary does not know the secret of any registered camera. Then, except with negligible probability, some registered camera produced a proof for a pair (V, m) with $m' = m$ and V' equal to V (or to its honest stego stream) outside the carrier set $\Gamma_K(V)$.*

Proof. Acceptance means a valid proof for (ρ, bind') with $\text{bind}' = \text{SHA256}(\text{dom} \parallel 0 \parallel \text{dig}_K(V') \parallel \text{SHA256}(m'))$. If no honest proof was ever produced for bind' , weak simulation-extractability yields a witness (s', path) whose leaf $\text{Poseidon}(s')$ opens to ρ ; by collision resistance of the tree the leaf is a registered key, and s' is then a preimage of an honest camera's public key, contradicting one-wayness. Hence an honest camera proved $\text{bind}' = \text{bind}$ for some (V, m) . Collision resistance of SHA-256 gives $\text{SHA256}(m') = \text{SHA256}(m)$, so $m' = m$, and $\text{dig}_K(V') = \text{dig}_K(V)$; Lemma 2 concludes. $\square$

Remark 1 (What the carriers can still hold). Theorem 1 leaves the carrier bits free, but those bits *are* the payload: they must parse as a frame whose message is m and whose proof verifies for bind . An adversary can therefore at most replace π by a re-randomized proof for the same statement. Flipping a carrier changes the extracted payload and is rejected (Section IX); flipping any other bit changes the digest.

Corollary 1 (Attacks covered). *Each of the following yields rejection: embedding a valid payload into another video (the digest of the other video differs); dropping, appending or truncating NAL units; editing any non-carrier bit, including a trailing-one sign in a carrier IDR or any bit of a later frame; re-encoding; replacing the message; a proof made by a camera outside the trusted registry (it opens to a different root).*

Proposition 3 (Unforgeability with the stego key, G3). *Holding K does not help produce an accepting proof for a new binding: K enters only the schedule, the whitening and the mask, all of which are public to the verifier anyway, while acceptance requires a proof for bind , which by the argument of Theorem 1 requires a registered secret.*

This is the property a MAC-based design lacks: with a tag under K , every verifier can authenticate arbitrary content.

Proposition 4 (Anonymity, G4). *The verifier's view— ρ , bind and π —has the same distribution for every registered camera.*

Proof. ρ and bind do not depend on which leaf is used. By perfect zero knowledge, π can be simulated from the statement alone, so its distribution does not depend on the witness. $\square$

Anonymity is relative to the registry: a registry of one camera identifies it.

Coverttness (G5). Without K the schedule is a pseudo-random subset of the candidates and the embedded bits are pseudo-random; with a wrong key the version byte matches with probability 2^{-8} and the length must additionally be admissible, after which the payload parser and the proof reject. Sign flips do not change the bit rate, the codeword lengths or the file size apart from rare emulation-prevention bytes. We only measure first-order statistics: the share of negative trailing ones does not change significantly (Section IX). Calibrated or learned steganalysis is future work, and we do not claim security against it.

Proposition 5 (Token scope). *In per-video key mode, a verification token t_V lets its holder extract and verify V , and gives no advantage in computing K or the token $t_{V'}$ of any video V' with $\nu(V') \neq \nu(V)$.*

Proof. ν depends only on syntax that embedding does not change, so the holder recomputes ν from the stego stream and t_V equals the subkeys the sender used; extraction and the digest of Algorithm 2 need nothing else. Tokens are outputs of HKDF-Expand (HMAC-SHA256, a PRF) under the secret k_ν , which is itself a PRF output under the extracted key of K on input ν . Tokens of different nonces are therefore computationally independent, and recovering K from t_V would invert HMAC. $\square$

Disclosing t_V deliberately gives up the coverttness of V (its carriers and plaintext become known to the token holder); the coverttness of every other video under the same K is preserved. Two videos with identical first-IDR syntax up to the signs would share a nonce; for natural content this does not happen.

Low-drift policy and coverttness. The policy keeps the order keyed inside each tier, but it concentrates carriers in low tiers (bottom of the picture, high-frequency AC). An adversary who knows the policy knows where carriers are more likely; whether this helps detection is an empirical question, measured in Section IX-F.

Mode 1. A live stream embeds its proof in the first IDRs, before the rest exists, so the proof can only bind the message.

Theorem 1 then holds for m but not for the video, and a mode-1 payload can be replayed into another stream. Verifiers reject mode 1 unless explicitly configured otherwise.

VII. A DISTORTION MODEL FOR TRAILING-ONE SIGN EMBEDDING

The forward direction of the channel—choose candidates, embed, measure PSNR—does not tell a designer how many bits a frame can take before its quality drops below a target. This section derives the distortion of one flip in closed form, models its propagation, and inverts the result into a per-IDR bit budget. All energies are sums of squared errors (SSE) of 8-bit samples; PSNR = $10 \log_{10}(255^2 N_{\text{px}}/\text{SSE})$ for a plane of N_{px} samples.

A. Direct distortion of one flip

Lemma 3 (Single flip). *Flipping the sign of a trailing one at raster position (i, j) of a 4×4 block coded with quantization parameter QP changes the reconstructed residual of that block, before rounding and clipping, by an error of energy*

$$E_{ij} = 4 Q_{\text{step}}(QP)^2 \kappa_m(i, j), \quad (18)$$

$$\kappa_m(i, j) = \frac{v_m(i, j)^2 \|b_i\|^2 \|b_j\|^2}{16 v_{m,0}^2}, \quad (19)$$

with $m = QP \bmod 6$. $\kappa_m(i, j) = 1$ for the even/even class (including the DC position) and $\kappa_m(i, j) \in [0.925, 1.057]$ for all positions and all m .

Proof. The level changes by $\Delta c = \mp 2$. By Eq. (6) the scaled coefficient changes by $\Delta d = \mp 2 v_m(i, j) 2^{\lfloor QP/6 \rfloor}$ and no other coefficient changes. By linearity of Eq. (3) before the final rounding, the residual error is $\Delta e(x, y) = \Delta d b_i(x) b_j(y)/64$, whose energy is $(\Delta d/64)^2 \|b_i\|^2 \|b_j\|^2$. Substituting $Q_{\text{step}} = v_{m,0} 2^{\lfloor QP/6 \rfloor}/16$ gives Eq. (18). The bounds follow by evaluating the six tables: $\kappa \in \{1, 1.056\}$, $\{1, 1.012, 1.046\}$, $\{0.925, 0.947, 1\}$, $\{1, 1.033, 1.054\}$, $\{0.954, 0.977, 1\}$ and $\{1, 1.014, 1.020\}$ for $m = 0, \dots, 5$. $\square$

The near-constancy of κ is not a coincidence: the scaling factors v_m were designed to compensate the unequal norms of the integer basis [36], so the scaled transform is close to orthonormal and, by Parseval, a level change of ± 2 costs $(2Q_{\text{step}})^2$ wherever it occurs. The energy depends on the quantizer only; it does not depend on the image, the prediction mode or the frequency.

Lemma 4 (DC paths). *A flip in an Intra16×16 LumaDC block or a ChromaDC block has, before rounding, the same total energy $4Q_{\text{step}}^2$ (with QP_c for chroma), spread uniformly over the sixteen (resp. four) 4×4 blocks of the MB component.*

Proof. LumaDC: the Hadamard output changes by ± 2 in all sixteen positions, and the DC scaling $(f \cdot 16 v_{m,0} 2^{\lfloor QP/6 \rfloor})/64$ turns this into a change of $v_{m,0} 2^{\lfloor QP/6 \rfloor}/2$ in the DC coefficient of each block, i.e. a constant residual change of $v_{m,0} 2^{\lfloor QP/6 \rfloor}/128$ over each block; the total energy is

$256 (v_{m,0} 2^{\lfloor QP/6 \rfloor}/128)^2 = 4Q_{\text{step}}^2$. ChromaDC: the 2×2 Hadamard output changes by ± 2 in four positions, the scaling $(f \cdot 16 v_{m,0} 2^{\lfloor QP_c/6 \rfloor})/32$ gives $v_{m,0} 2^{\lfloor QP_c/6 \rfloor}$ per block, and four blocks of sixteen samples give the same total. $\square$

Additivity. The schedule uses one candidate per block, so two flips in the same block can only be a DC-path flip and an AC flip of the same MB component. Their errors are orthogonal (an AC basis function has zero mean, a DC change is constant over the block), so the direct energies add exactly. Rounding by $(\cdot + 32) \gg 6$ adds a bounded perturbation; in the demo block of Appendix A the exact rounded energy is 128 against a model value of 121.

B. Propagation

Ignoring clipping and the rounding of prediction, intra decoding is affine in the residual: in decoding order every block is its prediction (a linear function of earlier reconstructed samples) plus its residual. A direct error δ_k of flip k therefore becomes a total error $\mu_k = (I - \mathcal{P})^{-1} \delta_k$, where $\mathcal{P}$ is the strictly lower-triangular prediction operator of the frame, and deblocking adds a further, data-dependent smoothing. We define the *propagation gain*

$$G_k = \frac{\|\mu_k\|^2}{4Q_{\text{step}}(QP_k)^2}, \quad (20)$$

which captures drift through intra prediction and deblocking ($G_k \approx \kappa_k$ when nothing propagates; rounding and clipping can push it slightly below). For a set of flips the total error is $\sum_k \mu_k$; cross terms $\langle \mu_k, \mu_l \rangle$ carry the signs of the flipped levels, which are those of the cover coefficients, and are zero-mean over the random keyed choice of carriers. Hence

$$\mathbb{E}[\text{SSE}] \approx \sum_k 4Q_{\text{step}}(QP_k)^2 G_k. \quad (21)$$

G_k depends on the prediction structure—MB type, prediction modes, block position, whether deblocking is on—which is why it is measured rather than derived (Section IX).

C. Forward model and inverse

With constant QP , N flips in the luma plane and a frame-level gain $\bar{G}$ (the geometric mean of G_k fitted on data),

$$\text{PSNR}_Y \approx 10 \log_{10} \frac{255^2 WH}{4 \bar{G} N Q_{\text{step}}(QP)^2}. \quad (22)$$

Three rules of thumb follow: doubling the number of flips costs 3.01 dB, raising QP by six costs 6.02 dB, and four times more pixels gain 6.02 dB at the same number of flips. Inverting Eq. (22) for a target T :

$$N_{\text{max}}(T) = \frac{255^2 WH 10^{-T/10}}{4 \bar{G} Q_{\text{step}}(QP)^2}, \quad \text{cap}(T) = \frac{N_{\text{max}}(T)}{p f_Y}, \quad (23)$$

where $p = 1/2$ is the probability that a whitened bit changes the sign and f_Y the share of luma candidates (a uniformly random schedule selects luma carriers at that rate). Chroma planes have their own budget with QP_c and the chroma gain.

Receiver computability. A per-IDR cap is usable only if the receiver derives the same value from the stego stream. Eq. (23) uses W , H , the slice QP, f_Y and a protocol constant $\bar{G}$; by Proposition 1 all of them are identical in the cover and the stego stream. An adaptive cap therefore needs no side information. The same constraint applies to any cost-aware schedule: costs may depend on positions, categories, MB types, prediction modes and coefficient magnitudes, but never on the sign values that carry data.

D. Towards minimal-distortion embedding

Eq. (18) supplies an additive per-carrier cost $\rho_k = 4Q_{\text{step}}(QP_k)^2\kappa_k\hat{G}_k$, with $\hat{G}_k$ predicted from invariant features (e.g. category and MB type). For a payload of L bits, the distortion-limited sender of [31], [32] changes carrier k with probability $p_k = 1/(1 + e^{\lambda\rho_k})$, where λ solves $\sum_k h(p_k) = L$ (h the binary entropy) and the expected distortion is $\sum_k p_k\rho_k$; syndrome-trellis codes come close to this bound. Even without costs, matrix embedding reduces flips per bit: the present channel spends 1/2 expected flip per bit, while a Hamming code with $r = 3$ embeds three bits into seven carriers with $1 - 2^{-3} = 0.875$ expected flips, i.e. 0.29 flips per bit, a $10\log_{10}(0.5/0.29) \approx 2.3$ dB gain at equal payload. Candidates are plentiful (thousands per IDR against 64 used), so the larger carrier count is affordable. We leave syndrome coding to future work. A simple integer version of the cost, the drift tier of Eq. (16), is implemented as the `low-drift` selection policy (Section V-H) and evaluated end to end in Section IX-E.

VIII. IMPLEMENTATION

Native codec layer (C++17). A self-contained H.264 Baseline parser splits Annex-B streams, removes emulation prevention, parses SPS, PPS and I-slices down to every CAVLC element, and lists candidates. The same library implements the segment protocol for files and for byte streams, the carrier-masked digest, and re-emission of patched NAL units with emulation prevention. Command-line tools read the 32-byte key from standard input, never from the command line: `embed-stream-auth-stdin`, `extract-stream-auth`, `video-digest`, live variants on `stdin/stdout`, and an inspection tool that prints NAL tables, per-block coefficients and per-segment candidates as JSON.

Proof layer. The circuit is written in Circom 2.1 with `circomlib`’s Poseidon; proofs are generated and verified with `snarkjs` 0.7.6 on Node.js 22. The setup script uses the public Hermès Powers of Tau of size 2^{16} [45], pinned by its published BLAKE2b hash, followed by two Phase-2 contributions with fresh entropy, a final beacon and a `zkey` `verify`. Both contributions run on one machine, so this is a demonstration ceremony; a deployment needs independent contributors. The verifier and the registry code (Python) reproduce the native digest byte for byte; a Python reference of the schedule and whitening is checked against the native codec in the test suite.

Service. A FastAPI application exposes asynchronous HTTP jobs (`embed`, `extract`, `verify`) and a WebSocket endpoint that pipes an Annex-B byte stream through the live embedder.

Keys and messages reach native processes only through standard input; job records store neither; artifacts are downloadable once. The `verify` job loads the registry root from configuration and accepts message-only proofs only when explicitly enabled.

Channel parameters. The selection policy and the key mode of Section V-H are command-line flags of every embedding, extraction and digest command (defaults reproduce the earlier protocol byte for byte), a `video-token` command prints the verification token of a video, and the inspection tool reports each candidate’s tier inputs so that the Python mirror can reproduce the native schedule exactly.

Testing. The test runner groups 161 checks into twelve phases (proof, native contract including Python–native equality of both policies and both key modes, CLI fixtures, VLC tables, service, HTTP/WebSocket, manifest security, scheduler, benchmark analysis including the distortion model, trust interfaces, demo arithmetic, demo smoke run), plus seven CTest cases for the native library (among them: per-video nonce identical for cover and stego, token extraction equal to master-key extraction, a token failing on another video). The evaluation code has its own tests, e.g. that each re-implemented baseline embeds exactly the message bits it claims. A terminal walkthrough performs 60 automated cross-checks, e.g. that FFmpeg’s decoder and our parser agree on every MB type and QP, that a Python re-implementation of dequantization, inverse transform and intra prediction reproduces FFmpeg’s samples, and that the Python and native digests agree.

IX. EVALUATION

A. Setup

All measurements ran on one laptop (Intel Core i7-12700H, 14 cores/20 threads, 25.4GB RAM, Windows 11) with FFmpeg 8.0.1 (x264 core 165), Node.js 22.20 and `snarkjs` 0.7.6. Sources are nine Xiph CIF sequences [46] (`akiyo`, `city`, `coastguard`, `container`, `deadline`, `football`, `two foreman` variants, `hall_monitor`), 30 frames each, encoded at their native 352×288 and Lanczos-upscaled to 640×480 and 1280×960 (upscaled sequences test scaling, not native high-resolution content). The benchmark encoder is x264 `ultrafast/zerolatency`, Constrained Baseline, QP 22 for P frames (the I-slice QP is 19), all-intra, one slice per IDR; this preset uses only 16×16 MBs and no deblocking. Each case embeds a 68-byte message with its proof (payload 201 B, 1,632 frame bits) at the default cap of 64 bits per IDR, i.e. into 26 of the 30 IDRs, under a fresh stego key and an 8-camera registry. Wall times include process start-up (Node.js for `snarkjs`).

B. Functional results

All 27 cases passed every acceptance check: native capacity scan, embedding, strict FFmpeg decoding (`-xerror`), blind extraction of the exact payload, a video-bound proof verified against the registry root with the digest recomputed from the stego file, rejection under a wrong key, and zero sign changes outside the keyed schedule (checked position by position). The stego file has exactly the size of the cover in all 27 cases. A

TABLE II

PER-FRAME QUALITY OVER THE 26 CARRYING FRAMES OF 9 SEQUENCES
PER RESOLUTION.

	CIF	640 × 480	1280 × 960
Cover→stego PSNR-Y, min [dB]	46.71	44.03	44.72
Cover→stego PSNR-Y, mean [dB]	57.44	59.69	62.64
Cover→stego SSIM, min	0.99871	0.99895	0.99960
Source→stego PSNR-Y, min [dB]	41.48	41.62	42.94
Source→stego PSNR-Y, mean [dB]	45.57	46.43	48.11
Changed signs per case (mean)	808	823	816
Candidates per case (mean)	149,374	420,193	1,287,323

separate end-to-end run on 300 CIF frames passed five of five trials.

C. Embedding quality

Table II reports PSNR-Y and SSIM [47] per frame, comparing the clean decode of the cover with the decode of the stego stream (incremental embedding distortion), and from the original source (encoding plus embedding). About half of the 1,632 scheduled signs change (808–823 on average), i.e. about 31 per carrying frame. No frame falls below the common 40 dB reference. The encoder’s own loss dominates: on average, coverstego PSNR is 12–15 dB higher than sourcestego PSNR.

D. Validating the distortion model

We test the model of Section VII in three experiments (scripts and raw data in the artifact): (B) single flips decoded in isolation, which measure the direct energy and the gain of individual carriers; (A) the 27 benchmark encodes, which measure frame-level distortion with ~ 22 luma flips per frame; and (C) an inverse check that sets the cap from a target PSNR, embeds and measures.

Experiment B: single flips. We encoded the first frame of *foreman* and *city* (CIF) with two x264 configurations—*ultrafast* (Intra16×16 only, deblocking off; the benchmark encoder) and *medium* (Intra4×4 and Intra16×16, deblocking on)—at QP 22, 28 and 34 (I-slice QP 19, 25, 31), sampled 100 candidates per encode (1,200 in total, skipping the rare carriers whose flip would add or remove an emulation-prevention byte), flipped each one alone and decoded with and without the loop filter.

Direct energy. For all 385 sampled Intra4×4 luma carriers, the change of the flipped block’s own samples in the unfiltered decode equals, sample for sample, the change of the residual computed from Eqs. (6) and (3) with rounding; the decoder adds nothing else to the block. The ratio of that exact energy to the model value $4Q_{\text{step}}^2$ has median 1.012 and mean 1.022 (5th–95th percentile 0.95–1.19; small residuals are the most affected by rounding), and the pre-rounding factor κ of the sampled positions lies in $[1.000, 1.056]$, as Lemma 3 states.

Gains. Table III summarizes G . Three observations matter for the model. First, G does not depend on QP (luma mean 6.6, 7.4, 6.0 for ultrafast and 38.4, 33.3, 34.0 for medium at the three QPs), so distortion scales as Q_{step}^2 , i.e. by 6 dB per six

TABLE III

SINGLE-FLIP PROPAGATION GAIN G BY CARRIER CATEGORY
(EXPERIMENT B; n FLIPS, MEAN, MEDIAN, 90TH PERCENTILE).

Category	n	ultrafast			n	medium		
		mean	med.	p90		mean	med.	p90
Luma AC / 4×4	352	3.7	1.04	5.7	395	35.4	2.1	52.9
LumaDC	38	33.8	4.9	64.4	5	16.3	12.6	31.6
ChromaAC	173	9.2	1.12	11.9	146	4.4	1.06	5.4
ChromaDC	37	23.9	7.1	65.3	54	30.5	11.4	102.2
All luma	390	6.6	1.05	7.8	400	35.2	2.2	47.5

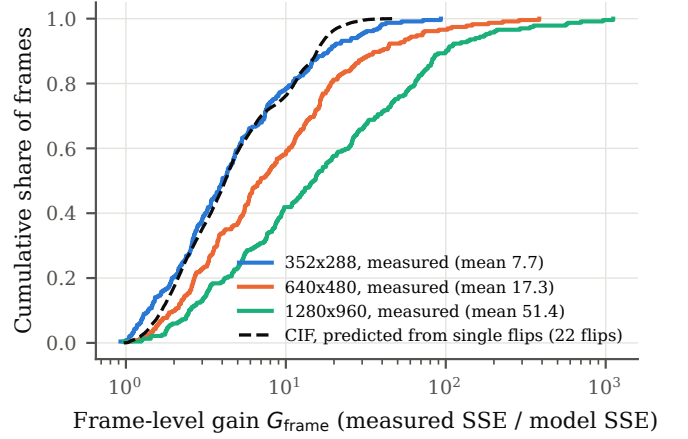


Fig. 2. Frame-level gain of the 702 benchmark frames with luma flips (experiment A), and its prediction for CIF from single flips alone (experiment B, 22 independent draws per frame).

QP steps. Second, the gain is heavy-tailed: the median carrier barely propagates ($G \approx 1$ –2), but single luma gains reach 309 (ultrafast) and 2,187 (medium), and the 10% largest gains account for 80% (ultrafast) and 84% (medium) of the total energy. Third, the encoder configuration matters more than anything else: the mean luma gain is 6.6 with Intra16×16 only and 35.2 with Intra4×4, whose nine directional modes chain many blocks. Deblocking changes the mean luma gain by about 1% (35.6 without, 35.2 with), so drift through intra prediction, not filtering, is the propagation mechanism.

Experiment A: sequences. For each of the 810 frames of the benchmark we located every changed bit in the bitstream, classified it with the parser, summed the model energy $4Q_{\text{step}}^2$ per plane and compared it with the measured SSE between the cover and stego decodes. Table IV reports the frame-level gain. Its *mean* (the ratio of total measured to total predicted energy) is the quantity Eq. (22) needs; for CIF it is 7.7, close to the single-flip mean of 6.6 measured on different frames in experiment B. More strikingly, Fig. 2 shows that drawing 22 single-flip gains at random reproduces the whole distribution of the CIF frame gains up to its 90th percentile (median 4.17 predicted vs. 4.00 measured); the measured upper tail is heavier (95th percentile 28.6 vs. 18.4), because carriers of the same frame share its content and are not independent. The

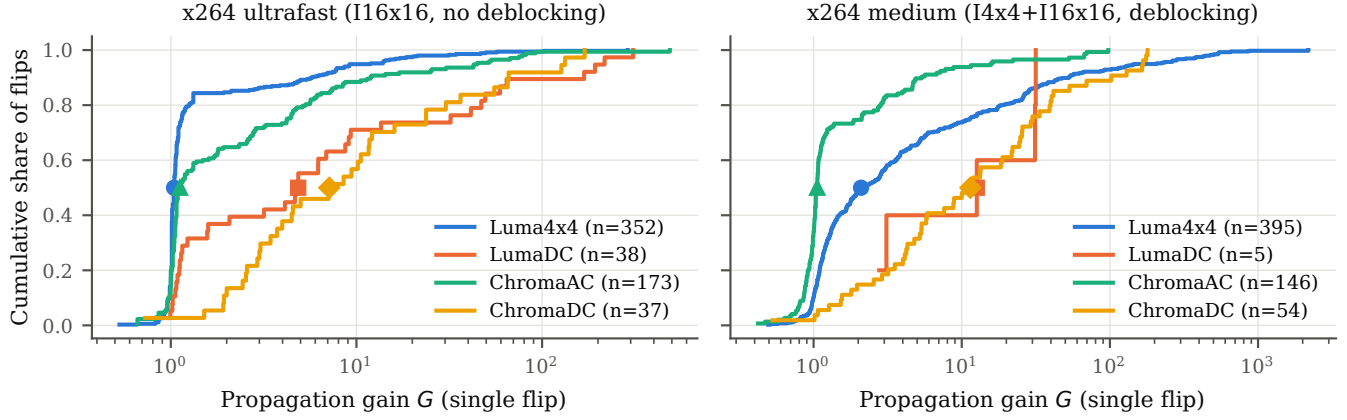


Fig. 3. Distribution of single-flip gains per carrier category for the two encoder configurations (markers: medians).

TABLE IV
FRAME-LEVEL GAIN IN THE 27-ENCODE BENCHMARK (EXPERIMENT A, ULTRAFAST, 64 BITS PER IDR).

	CIF	640 × 480	1280 × 960
Frames with luma flips	234	234	234
Luma flips per frame	22.0	22.7	22.9
Mean gain $\bar{G}$	7.7	17.3	51.4
Frame gain, median	4.0	7.0	15.2
Frame gain, 95th pct.	28.6	65.8	173.8
Measured PSNR-Y, median [dB]	57.9	60.1	62.7
Per-frame error, MAE [dB]	4.1	5.1	6.7

gain grows with resolution (7.7, 17.3, 51.4). Our upscaled sequences contain large smooth regions, where Intra16×16 prediction carries a change across many MBs; native high-resolution content may behave differently, and calibrating $\bar{G}$ per resolution and encoder is necessary.

Because a frame’s distortion is dominated by its few high-gain carriers, the model predicts the *distribution* of frame PSNR, not individual frames: with the per-resolution mean gain, the per-frame PSNR error has a mean absolute value of 4.1–6.7 dB and a correlation of only 0.19–0.35 with the measurement (leave-one-sequence-out errors are the same to 0.05 dB, so the gain does not overfit a sequence).

Experiment C: the inverse. We encoded 30 frames of *foreman* (CIF) with both configurations, computed the cap with Eq. (23) for $T = 40, 45$ and 50 dB under three choices of the gain, embedded a random payload filling the capacity, and repeated each setting with five independent keys and payloads (145 fully loaded frames per setting): (i) *expected*, the mean single-flip gain of experiment B, which makes the expected frame energy equal to the budget; (ii) *p95-MC*, the 95th percentile of the frame gain obtained by Monte Carlo from single flips; (iii) *p95-sequence* (ultrafast only), the 95th percentile of the measured CIF frame gains of experiment A. The single-flip gains are pooled over both sources and all three QPs, which Table III justifies for QP (the QP-19 subset has

TABLE V
INVERSE CHECK (EXPERIMENT C): GAIN USED, CAP, SHARE OF FRAMES WITH PSNR-Y $\geq T$ AND 5TH-PERCENTILE PSNR-Y [dB]; ^s: CAP AT OR ABOVE THE CANDIDATE COUNT OF AT LEAST 97% OF THE FRAMES’ IDRS (SATURATED).

Encoder	Policy	G	T	cap	$\geq T$	p05
medium	expected	35.2	45	200	61%	40.2
	expected	35.2	50	63	70%	45.1
	p95-MC	63.4	40	351	86%	38.3
	p95-MC	102.4	45	68	88%	43.2
	p95-MC	140.1	50	15	86%	43.6
	expected	35.2	40	634 ^s	75%	37.3
ultrafast	expected	6.6	50	311	46%	44.3
	p95-MC	15.8	50	130	80%	47.4
	p95-seq.	28.6	45	228	92%	43.7
	p95-seq.	28.6	50	72	91%	48.8
	all three	–	40	≥ 721 ^s	99%	41.5

mean luma gain 6.6 and 38.4 against 6.6 and 35.2 pooled); the pool contains the first frame of *foreman* but none of the other 29 test frames. The third policy uses data that include these frames’ sequence. A cap at or above the number of candidates in an IDR (about 650 for ultrafast and 460 for medium here) saturates: every candidate is used and the target no longer controls anything.

Table V and Fig. 4 give the outcome. The *expected* policy places the median frame near the target, as intended (46–70% of frames reach T). A percentile policy trades capacity for reliability: with the Monte Carlo percentile 80–88% of frames reach T , short of the nominal 95% for the reason seen in Fig. 2 (correlated carriers within a frame), while the percentile measured on sequences reaches 91–92%. Content is the other factor: the single-flip mean luma gain of *foreman* alone at QP 19 is about twice the pooled value (12.9 vs. 6.6 for ultrafast, 70 vs. 35 for medium, $n \approx 48$ each), so a gain calibrated on other content under-protects a sequence that propagates more. In every non-saturated setting the 5th percentile stays within 6.5 dB below the target, and the cap

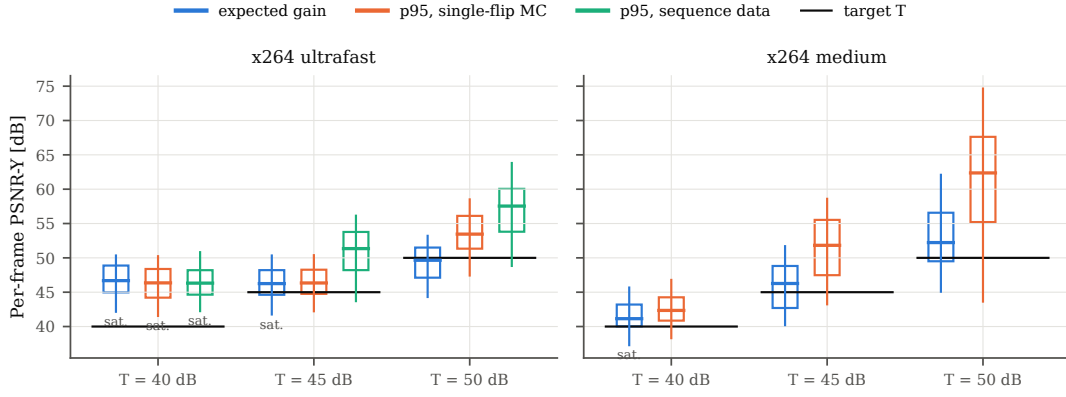


Fig. 4. Per-frame PSNR-Y for caps derived from a target T (boxes: quartiles; whiskers: 5th–95th percentiles; 145 frames per box).

moves in the direction and by the amount the model predicts: from $T = 45$ to 50 dB the cap shrinks by $10^{0.5} \approx 3.2\times$ under the expected policy (200 to 63, 983 to 311). Saturation itself is informative: at the benchmark’s QP the ultrafast stream cannot drop below about 41.5 dB at its 5th percentile even when every candidate is used.

What the data suggest for carrier selection. Since a tenth of the carriers produces four fifths of the distortion, excluding high-gain carriers is worth more than reducing the payload. The single-flip data show which invariant features predict the gain. In the medium configuration, low-frequency trailing ones (coefficient positions with $i+j \leq 1$) have mean gains of 84–92 against 1.6 at $i+j = 6$, because low-frequency basis functions change the right column and bottom row from which later blocks predict. In both configurations, carriers in the top third of the picture have mean gains of 13.9 (ultrafast) and 65.6 (medium), against 2.2 and 6.1 in the bottom third, because more blocks are predicted from them. Restricting a random schedule to bottom-third carriers would reduce the expected energy by 4.8 dB (ultrafast) and 7.6 dB (medium) according to these averages, and excluding DC paths by 2.5 dB (ultrafast). These are projections from single-flip averages, not end-to-end measurements; the cost-aware schedule they motivate keeps the receiver-computability constraint of Section VII.

E. Comparison with published sign-domain methods

Material. Besides the eight CIF sequences we use 19 native high-resolution, camera-captured Xiph sequences [46] (twelve at 1920×1080 : crowd_run, ducks_take_off, in_to_tree, old_town_cross, park_joy, pedestrian_area, riverbed, rush_hour, station2, sunflower, tractor, blue_sky; seven at 1280×720 : mobcal, parkrun, shields, stockholm and the three vidyo conferencing clips), 60 frames each. All covers are encoded with x264 medium (Intra4×4 and Intra16×16, deblocking on), Constrained Baseline, QP 18, 22, 28 and 34, with GOP 1 (all intra) and GOP 30 (one IDR followed by P frames, so that drift into predicted frames is included).

Methods. All methods use the same channel and are fed random message bits: per IDR they place N bits, $N = 64$ (the

system default) or $N = 5\%$ of the IDR’s candidates. Besides our two policies we re-implement the selection and mapping rules of four published trailing-one methods inside a sign-only channel: Kim *et al.* [16] (first trailing-one sign of luma blocks in decoding order), Lin *et al.* [17] (keyed luma blocks, XOR of all trailing-one signs), Wang and Ma [18] (one host per E candidates, $(1, 3, 2)$ matrix code; the paper fixes $E \in [12, 20]$; we choose E from the expected bits per host so that the hosts span the whole picture at the given payload), and a Liao-style per-macroblock trailing-one parity [19], [20] (the original grouping is not fully specified in the sources we could access). Ma *et al.* [21] modify coefficient pairs and need neighbouring prediction modes, so it is not reducible to sign flips and is not included; codeword-substitution methods [14] are not reducible either. Because only the selection and mapping rules carry over, our numbers are not comparable with those reported in the original papers. Unit tests check that every method embeds exactly the message bits it claims.

Quality. Table VI and Fig. 5 show the same ordering at every QP, GOP and resolution. The low-drift policy is best: over the 16 settings (4 QPs, 2 GOPs, 2 rates) it gains 3.3–7.2 dB over the uniform keyed choice on CIF (median 4.9 dB; e.g. 53.3 vs. 49.3 dB at QP 22 and 40.9 vs. 35.2 dB at QP 34, all intra) and 3.3–8.0 dB on native high resolution (median 5.9 dB; 61.5 vs. 54.4 dB at QP 22), and 3.9–9.2 dB over the best baseline of each setting. The uniform keyed choice is in turn better than every baseline, because spreading carriers over the picture avoids the long intra-prediction chains that start at the top of the frame; the methods that fill blocks in decoding order are worst (Kim *et al.*: 37.7 dB at QP 22, 24.2 dB at QP 34 on CIF). Parity and matrix codes change fewer signs per bit (Wang and Ma: 0.44 changes per bit against 0.50) but do not compensate for where they write: these methods use luma blocks only, whose sign changes propagate furthest with Intra4×4 prediction (mean single-flip gain 35 for luma against 4 for chroma AC under this preset, Table III), and parity and matrix codes may flip a lower-frequency trailing one than the first. No method changes the bit rate; the file size changed by one byte in a single run (an emulation-prevention byte).

TABLE VI
EMBEDDING QUALITY OF THE COMPARED METHODS AT 64 BITS PER IDR: MEDIAN OVER SEQUENCES OF PSNR-Y OVER ALL FRAMES [dB] (X264 MEDIUM). GOP 30 INCLUDES THE DRIFT INTO 29 P FRAMES PER IDR.

Method	CIF (8 sequences)				native 720p/1080p (19 sequences)			
	GOP 1		GOP 30		GOP 1		GOP 30	
	QP 22	QP 34	QP 22	QP 34	QP 22	QP 34	QP 22	QP 34
Ours, low-drift	53.3	40.9	53.8	41.5	61.5	47.5	62.3	49.8
Ours, random	49.3	35.2	49.1	38.2	54.4	41.0	56.6	44.9
Wang & Ma [18]	45.2	31.4	44.7	31.7	45.0	31.2	48.9	34.5
Lin et al. [17]	46.9	33.6	49.1	34.4	52.7	38.7	54.9	40.6
Liao-style [20]	44.1	29.4	45.2	30.2	39.3	26.0	42.2	29.3
Kim et al. [16]	37.7	24.2	36.3	25.6	35.1	22.9	37.4	26.8

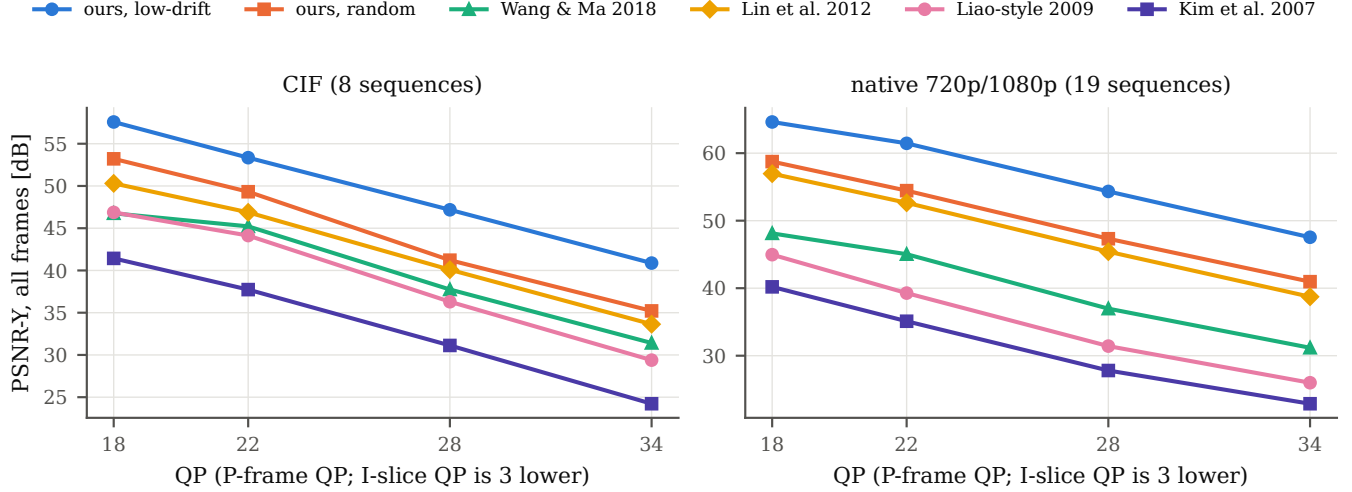


Fig. 5. PSNR-Y (all frames, median over sequences) against QP at 64 bits per IDR, all-intra.

F. Steganalysis

Protocol. Covers are x264 medium, QP 22, all-intra encodes (CIF: 60 frames; high resolution: 10 frames, since feature extraction at 1080p is the bottleneck). Each of the six methods embeds at three rates: 64 bits per IDR, 5% and 20% of the IDR’s candidates, giving 486 cover/stego pairs. The 27 sources are split by sequence into 17 for training, 3 for validation and 7 for testing (2 CIF and 5 high-resolution sequences, 160 frames per class), so no content is shared between training and testing. We use four detectors: (i) 323 compressed-domain features of the trailing-one signs (sign shares per category and coefficient class, joint sign patterns of a block, co-occurrences with the next level and with neighbouring blocks); (ii) SPAM-686 [48]; (iii) a 3,978-dimensional reduced rich model (“SRM-lite”) following [49]; features (i)–(iii) are classified with the FLD ensemble [50]; and (iv) a Xu-Net-style CNN [51] trained on 256×256 luma tiles with early stopping on the validation sources. We report the minimum average error $P_E = \min(P_{FA} + P_{MD})/2$ on the test frames, the minimum taken over all thresholds of the test ROC as is customary [50]; this favours the detector (it is not a threshold fixed in advance), so it is conservative for claims of undetectability. With 320 test frames its standard error is about 0.03, so differences below

0.05 are not significant.

Results. At the system’s operating point of 64 bits per IDR no detector does better than guessing for any method (P_E 0.44–0.49; AUC 0.50–0.55): about 32 sign changes among thousands of candidates per frame leave no measurable trace for these detectors. At 5% of the candidates P_E falls to 0.33–0.44 and at 20% to 0.17–0.42 (Table VII, Fig. 6). The compressed-domain features are the strongest detector of the method that fills luma blocks in decoding order (Kim *et al.*: $P_E = 0.17$ at 20%), whose changes concentrate in one region and one category. The low-drift policy is not easier to detect than the uniform keyed choice with the pixel-domain detectors (SPAM 0.36 vs. 0.36, SRM-lite 0.33 vs. 0.32) and slightly easier with the CAVLC features and the CNN at 20% (0.25 vs. 0.29 and 0.25 vs. 0.28), differences within about one and a half standard errors. Concentrating carriers in cheap regions therefore buys 3–8 dB of quality without a measurable loss of security at the rates the system uses; at high rates a small loss cannot be excluded.

These detectors are a lower bound on what a determined warden can do: the training set has 17 sequences, and stronger models (the full 34,671-dimensional SRM, SRNet-class networks, detectors that exploit the drift pattern of trailing-one changes [23]) and larger training sets may detect lower rates.

TABLE VII
DETECTION ERROR P_E ON THE TEST SOURCES (0.5 = RANDOM GUESSING) PER DETECTOR AND EMBEDDING RATE: 64 BITS PER IDR (SYSTEM DEFAULT), 5% AND 20% OF THE IDR'S CANDIDATES.

Method	CAVLC features			SPAM-686			SRM-lite			CNN		
	64	5%	20%	64	5%	20%	64	5%	20%	64	5%	20%
Ours, low-drift	0.47	0.42	0.25	0.48	0.44	0.36	0.48	0.37	0.33	0.48	0.43	0.25
Ours, random	0.47	0.43	0.29	0.47	0.44	0.36	0.46	0.40	0.32	0.48	0.41	0.28
Wang & Ma [18]	0.47	0.43	0.32	0.47	0.42	0.32	0.46	0.39	0.28	0.46	0.33	0.22
Lin et al. [17]	0.47	0.43	0.29	0.48	0.41	0.31	0.46	0.36	0.33	0.46	0.38	0.30
Liao-style [20]	0.44	0.43	0.37	0.47	0.40	0.42	0.44	0.38	0.32	0.49	0.40	0.34
Kim et al. [16]	0.48	0.37	0.17	0.47	0.44	0.34	0.46	0.39	0.31	0.48	0.42	0.28

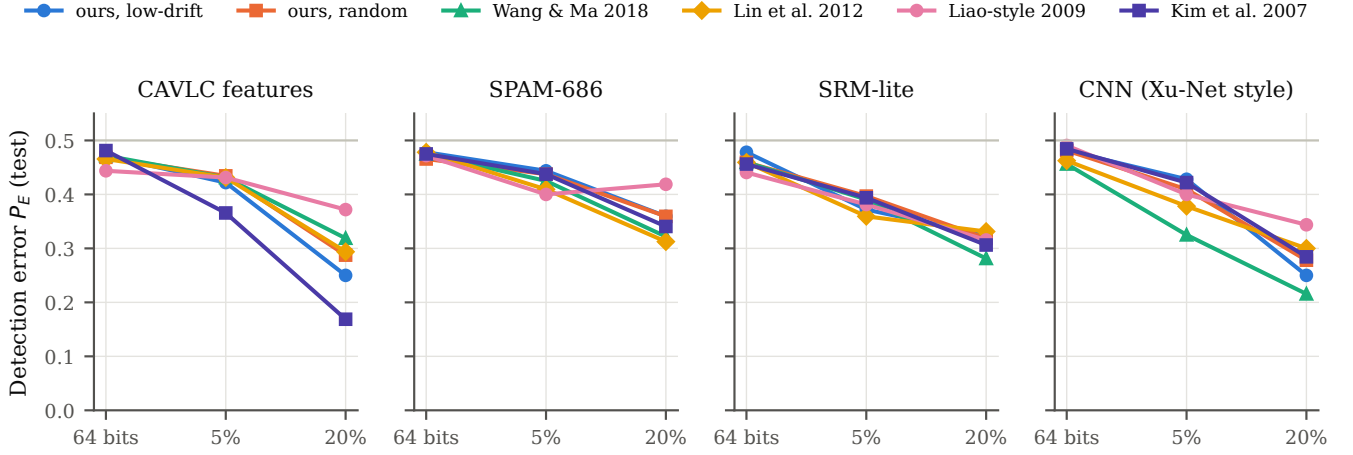


Fig. 6. Detection error P_E against the embedding rate for the four detectors.

We therefore claim covertness only at the default rate and only against the detectors tested.

G. Native low-drift policy and verification tokens

Table VIII runs the deployed native codec with a 201-byte payload (68-byte message and proof) at 64 bits per IDR on the 8 CIF and 19 native high-resolution covers (x264 medium, QP 22, all intra, 60 frames) for the four parameter combinations. In all 108 runs the master key extracts the exact payload and the stego file has the size of the cover. The low-drift policy raises the median PSNR-Y of the carrying frames by 3.9dB on CIF and 6.6dB on high resolution (61.1 vs. 54.5 dB), and the worst frame by 6.4 and 10.8 dB, at the same embedding time. In per-video key mode the token of each video extracts the payload and yields the same video digest as the master key in all 27 videos; the token of one video fails the frame check (“version is invalid”) on the next video in all 25 pairs tested; computing a token takes 0.07 s at CIF and 0.7 s at 1080p (one parse of the first IDR). Per-video keys add no measurable cost to embedding.

H. Sign statistics

Over all candidate signs of a case, the share of negative trailing ones is 0.504 before and after embedding; the two-proportion z -statistic is at most 0.223 in absolute value over

TABLE VIII
NATIVE CODEC WITH BOTH CHANNEL PARAMETERS (8 CIF AND 19 NATIVE HD COVERS, 60 FRAMES, 201-BYTE PAYLOAD, 64 BITS PER IDR): RUNS, EXTRACTION WITH THE KEY AND WITH THE TOKEN, PSNR-Y OF CARRYING FRAMES (MEDIAN, MINIMUM) AND EMBEDDING TIME [S].

Select	Key mode	n	key	token	PSNR	min	time
CIF random	master	8	8/8	–	49.4	38.7	1.4
CIF random	per-video	8	8/8	8/8	49.0	37.9	1.3
CIF low-drift	master	8	8/8	–	53.3	45.2	1.3
CIF low-drift	per-video	8	8/8	8/8	53.6	44.9	1.3
HD random	master	19	19/19	–	54.5	38.0	36.5
HD random	per-video	19	19/19	19/19	54.7	39.4	36.8
HD low-drift	master	19	19/19	–	61.1	48.8	33.6
HD low-drift	per-video	19	19/19	19/19	61.6	49.7	33.8

the 27 cases, far from the 1.96 threshold of a 5% test. This is a first-order indicator only (Section X).

I. Performance

Table IX compares Groth16 and PLONK on the same R1CS and witness; both verify a correct proof and reject a tampered public input. Groth16’s 129-byte proof is what makes the covert channel practical: snarkjs serializes a PLONK proof as 2.25 KB of JSON, which would need about 290 IDRs at the default cap, and even a compact binary encoding is several times larger than 129 B. Table X gives the end-to-

TABLE IX

PROOF SYSTEMS ON THE CAMERA_VIDEO CIRCUIT (8,737 CONSTRAINTS, 3 PUBLIC INPUTS).

	Groth16	PLONK
Trials	5	3
Prove [s]	1.52–1.60	21.5–23.3
Verify [s]	0.91–1.01	0.90–0.94
Proof size	129 B (compressed)	2.25 KB (JSON)
Peak prover RSS [MB]	695	864
Setup	circuit-specific (Phase 2)	universal SRS, 3.2 s
Proving key	4.2 MB	30.1 MB

TABLE X

END-TO-END STAGES, 300 CIF FRAMES, 5 TRIALS (MEDIAN, MS).

	Sender		Receiver	
Cover digest	511	Extract	535	
Groth16 prove	1,774	Digest + verify	1,448	
Embed	500	Strict decode	150	
Total	2,779	Total	1,989	

end stages for 300 CIF frames. Native costs scale with the number of parsed IDRs: the median capacity scan takes 52, 220 and 770 ms per IDR at the three resolutions, embedding 61, 245 and 891 ms per carrying IDR, and the digest, which parses the 26 carrying IDRs, 1.6, 6.4 and 22.6 s. Parsing, not proving, bounds throughput at high resolution; the parser is single-threaded and unoptimized.

J. Attack matrix

Table XI lists the attacks run through the service’s verify job on the end-to-end stream. Only the untouched stream is accepted. Notably, flipping a sign that does not carry the frame—in an IDR that does carry it—is rejected through the digest, and so is the same valid payload embedded into another video; a camera that builds its own registry produces a proof that verifies against its own root but not against the trusted one.

X. DISCUSSION AND LIMITATIONS

Where the design fits. The scheme protects bitstreams that are moved without re-encoding: surveillance and body-camera evidence chains that archive the camera’s H.264, files exchanged as files, and newsroom or legal workflows that need to show that footage comes from an authorized device without naming it. In these settings metadata is routinely lost while the elementary stream survives, and hiding the proof keeps an adversary from selectively removing it.

Why not a SEI message? A proof could be carried in a user-data SEI NAL unit at no capacity cost and with the same digest construction (the SEI unit would be excluded instead of the carrier bits). The difference is covertness: a SEI payload is visible to every parser and trivially stripped by a sanitizer, while the sign channel is located only with K . When covertness is not needed, the SEI variant is simpler, and our binding and proof carry over unchanged.

TABLE XI

ATTACK MATRIX (VERIFY JOB OUTCOME AND REASON).

Case	Result (rejection reason)
Untouched stego stream	accepted
Wrong stego key	payload_not_found
Flip a scheduled carrier sign	malformed_proof_payload
Flip non-carrier sign, carrier IDR	proof_invalid
Flip a sign in a later frame	proof_invalid
Drop the first IDR	payload_not_found
Truncate to half	proof_invalid
Re-encode (transcode)	payload_not_found
Swap the message	proof_invalid
Camera outside the registry	proof_invalid
Replay into another video	proof_invalid

Who can verify. Extraction and the carrier mask depend on the stego key, so by default verification is limited to key holders, who still cannot forge proofs (G3). Per-video key mode (Section V-H) makes verification of a chosen video public: its 64-byte token is handed to a court, a platform or published next to the file, and only that video loses its covertness. Fully keyless verification of every video would require the carrier set to be public, which contradicts hiding the payload.

Fragility. Any re-encode, resize or transcode destroys the payload and changes the digest. Robust provenance across edits requires transformation proofs [4], [6], [8] or a robust watermark acting as a soft binding [1]; combining such a layer with an in-band anonymous proof is an open direction.

What the proof does not show. A registered camera can film a screen; the camera secret lives in software in our prototype, so a compromised device can sign anything; registry governance (enrollment, revocation by publishing a new root, root distribution) is outside the system. Binding the secret to a secure element or an attested sensor pipeline, as in ZK-IMG [5], would address part of this.

Live streams. In mode 1 the proof binds the message only and can be replayed; video binding for live streams would need a proof that commits to frames embedded later, e.g. a chain of proofs or a final proof carried in a trailing segment.

Codec scope. The channel needs CAVLC and IDR frames. Most capture devices use the High profile with CABAC, where sign bits are bypass-coded but not length-invariant in the same simple way; HEVC uses sign data hiding, which interacts with sign embedding. Porting the channel is engineering work with its own distortion and syntax analysis.

Trusted setup. Groth16 needs a circuit-specific ceremony; our ceremony is a demonstration. PLONK removes the circuit-specific phase at a cost of roughly $14\times$ in proving time and a 2.25 KB proof (Section IX), which no longer fits the covert channel at the default cap.

Steganalysis. Four detectors (compressed-domain features, SPAM, a reduced rich model, a CNN) are at chance at the default rate and succeed at 20% of the candidates (Section IX-F). This bounds what moderately strong detectors trained on 17 sequences achieve, not what a determined warden with the

full rich model, modern networks and large training sets could do [52]; covertness claims should stay tied to the default rate. Minimal-distortion coding with the costs of Section VII would reduce the number of changes and is the natural next defense.

XI. CONCLUSION

We showed that a zero-knowledge proof of anonymous camera membership is small enough to be hidden in the trailing-one signs of an H.264 stream, and that it can be bound to the exact stream that carries it through a digest that masks only the keyed carrier positions. The resulting system needs no metadata, hides the signer within a registry, rejects replay, truncation, edits and re-encodes, and lets a verifier who holds the stego key check a proof that it cannot forge. We also derived a closed-form distortion model of sign embedding—one flip costs $4Q_{\text{step}}^2$ up to a factor in $[0.925, 1.057]$ —and turned it into a per-frame bit budget for a target PSNR, which our experiments confirm once the propagation gain of the encoder configuration is known. A drift-tier selection policy derived from the model gains 3–8 dB over a uniform keyed schedule and more over four re-implemented published methods, on CIF and on native 720p/1080p material, without a measurable loss of detectability at the operating rate; per-video keys let a single video be verified publicly through a token without exposing the stego key. Future work includes minimal-distortion coding with the derived costs, stronger steganalysis, CABAC and HEVC, hardware-bound camera secrets and video binding for live streams.

ARTIFACT AVAILABILITY

Source code, circuit, benchmark scripts and the recorded results used in this paper are available at <https://github.com/Saudadeeee/zk-snark-X-Steganography>.

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APPENDIX A

WORKED EXAMPLE OF ONE CARRIER

We follow one carrier of a demonstration run (`akiyo`, CIF, x264 default preset, I-slice QP 19, so $Q_{\text{step}} = 5.5$ and $4Q_{\text{step}}^2 = 121$) from the bitstream to the pixels. The carrier is the first trailing-one sign of luma block 5 of MB 212 (Intra4×4, prediction mode Diagonal_Down_Right), at RBSP bit 57,561 of the first IDR NAL unit.

CAVLC. With $n_C = 6$ the block is coded as `coeff_token=1011` (TotalCoeff 4, TrailingOnes 3), sign flags `011`, one level 1, `total_zeros=111` and two `run_before` codes. The scheduled bit is the first sign flag; whitening requires a 1, so the flag becomes 1 and the flags read `111`. All other bits keep their position and meaning. In the file the byte at offset 7,805 changes from `bf` to `ff`.

Levels and scaling. The scan $(1, -1, 0, -1, 1, 0, \dots)$ becomes $(1, -1, 0, -1, -1, 0, \dots)$: zig-zag position 4, raster $(1, 1)$, an odd/odd coefficient. With $v_1(1, 1) = 18$ and $2^{\lfloor 19/6 \rfloor} = 8$ the scaled coefficient changes from 144 to −144:

$$D = \begin{bmatrix} 88 & -112 & 0 & 0 \\ 0 & \pm 144 & 0 & 0 \\ -88 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Residual. The inverse transform gives

$$\hat{e}_{\text{before}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 2 & 3 & 3 \\ 0 & 1 & 4 & 6 \\ -4 & -2 & 2 & 4 \end{bmatrix}, \quad \hat{e}_{\text{after}} = \begin{bmatrix} -4 & -2 & 2 & 4 \\ 0 & 1 & 4 & 6 \\ 2 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 \end{bmatrix},$$

a change of energy $\sum \Delta \hat{e}^2 = 128$. Lemma 3 predicts $4Q_{\text{step}}^2 \kappa_1(1, 1) = 121 \times 1.046 = 126.6$ before rounding.

Reconstruction. A flip does not affect blocks decoded earlier, so if this were the only carrier the block’s prediction would be unchanged and its samples would change by exactly $\Delta \hat{e}$ (Experiment B confirms this for every sampled Intra4×4 block). In the demo a block decoded earlier also carried a flip, which shows up as a uniform change of +1 in this block’s prediction: an instance of intra drift. Our Python reconstruction (prediction plus residual) matches FFmpeg sample by sample before and after embedding, decoded with `-skip_loop_filter all`; deblocking then smooths the block edges.

APPENDIX B

NOTATION

K, s	stego key, camera secret
$k_{\text{sch}}, k_{\text{wh}}$	schedule and whitening subkeys
ρ, pk	registry root, camera public key Poseidon(s)
$\Gamma_K(V)$	carrier positions of stream V under K
$\text{dig}_K, \text{bind}$	carrier-masked digest, binding
F, P, π	frame, payload, compressed proof
cap	bits per IDR segment
$Q_{\text{step}}, v_m(i, j)$	quantizer step, scaling factor
κ, G	position factor, propagation gain
